Project Initialization and Planning Phase

Date	15 March 2024
Team ID	unm293200909
Project Title	FitBuddy - AI Fitness Plan Generator using Gemini Models
Name	Kavyasri.A
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to simplify fitness planning using generative AI. FitBuddy creates personalised 7-day workout plans, updates them from user feedback, and gives nutrition or recovery tips using Google Gemini models. Key features include AI-generated plans, a feedback loop and an admin dashboard for coaches.

Project Overview	
Objective	The primary objective is to help users get a personalised, goal-based 7-day workout plan instantly, and to refine it using feedback, with the help of Google Gemini AI models.
Scope	A FastAPI web application with an HTML (Jinja2) interface and API endpoints. It covers plan generation, plan updates from feedback, nutrition or recovery tips, SQLite storage and an admin dashboard.
Problem Statement	
Description	Beginners and busy people struggle to plan effective workouts. Personal trainers are costly, and generic online or app-based plans are not tailored to a person's age, weight, goal or intensity and cannot adapt to feedback.
Impact	Solving this gives users structured, affordable and personalised fitness guidance, improves motivation and consistency, and lets coaches monitor many clients from one place.
Proposed Solution	
Approach	Collect user details (name, age, weight, goal, intensity) and use Gemini 1.5 Pro to generate and update the 7-day plan, and Gemini Flash to generate quick nutrition or recovery tips. Store all data in SQLite through SQLAlchemy.

Key Features	- AI-generated personalised 7-day workout plan. - Feedback-based plan update (e.g. more cardio, more rest days). - Goal-based nutrition or recovery tip. - Admin dashboard to view all users and their plans. - Stored user data and plans in SQLite.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop CPU (dual-core or above); no GPU needed as AI runs through the Gemini API
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD (the app and SQLite database need only a small part)
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	SQLAlchemy, python-multipart, google-generativeai
Development Environment	IDE	VS Code / PyCharm, Python virtual environment, Git
Data		
Data	Source, size, format	User inputs (name, age, weight, goal, intensity) stored in SQLite; plans and tips generated by Gemini API as text

Work Assignment

Milestone	**Work Assigned to Kavyasri.A**
1. Model selection and architecture	Select Gemini models, define architecture, set up environment.
2. Core functionalities	Build plan generation, feedback update, tips and FastAPI backend.
3. routes.py development	Write main application logic and database handling.
4. Frontend development	Design index.html, result.html and all_users.html with Jinja2.
5. Deployment	Run locally with Uvicorn and test all scenarios.